

Ruturaj Mane

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PROFESSIONAL SUMMARY

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AI Engineer with strong analytical skills and hands-on experience in deep learning and intelligent systems. I work with complex datasets to build meaningful solutions, create reliable generative AI models, and improve their performance through model optimization and prompt tuning. I design and deploy scalable AI platforms that turn data insights into smart, efficient systems ready for real-world business use today.

WORK EXPERIENCE

DATA ANALYST INTERN, SmartBridge

07/2025 – 10/2025

- Extracted, cleaned, and transformed large datasets using SQL (PostgreSQL) and Python (Pandas, NumPy) to build analysis-ready datasets, improving data accuracy and reliability by 18%.
- Conducted exploratory data analysis, anomaly detection, and performance analytics to identify trends, risks, and data-quality gaps, leading to a 35% improvement in reporting effectiveness.
- Designed KPI-driven Tableau dashboards in collaboration with cross-functional teams, ensuring data validation, consistency, and audit-ready analytics outputs.

SKILLS SUMMARY

Programming & Development Skills Python, SQL, Git, APIs, GenAI, Statistics	Libraries & Frameworks Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch,
AI / Machine Learning Skills Machine Learning, Deep Learning, NLP, Computer Vision, Supervised & Unsupervised Learning, Model Evaluation	Data & Deployment Skills EDA, Data Cleaning, Feature Engineering, Dashboarding, Model Deployment, Cloud Basics, Prompt Engineering

Projects

NEET 2024 Result Analysis & Prediction 

08/2025

| Python(pandas, numpy, scikit-learn), Machine learning, Tableau

- Analyzed NEET 2024 data from over 1M+ records to find performance patterns and regional trends across 29 states. This generated insights that are useful for educational decision-making.
- Processed large datasets using Pandas and NumPy. I applied Scikit-learn models, including K-Means and Random Forest, for trend prediction and segmentation. This improved overall analytical accuracy by 35%.
- Integrated predictive insights into Tableau stories to compare actual and predicted outcomes. This allowed for clear reporting, risk identification, and data-driven interpretation.

AI Content Detector — Text Detection Module 

03/2026

| Python (PyTorch), RoBERTa, HuggingFace Transformers, Deep Learning, NLP

- Fine-tuned RoBERTa-base transformer model on a custom dataset of 442,000+ samples combining student essays, news articles, and Reddit conversations to detect AI-generated vs human-written text.
- Built an end-to-end data pipeline combining 6 real-world datasets, performed EDA, removed 211,000+ duplicates, and balanced classes to achieve unbiased model training across diverse writing styles.
- Achieved 98.65% validation accuracy on unseen test data. Identified & resolved training distribution bias by iteratively expanding dataset with improving real-world generalization significantly across casual, formal and conversational text.

Deep Learning–Based Road Segmentation from Satellite Imagery 

01/2026

| Python (PyTorch), U-Net, Deep Learning, Computer Vision, Streamlit

- Developed an automated road extraction system using binary semantic segmentation on over 1,100 high-resolution satellite image-mask pairs. Implemented a 4-level encoder-decoder U-Net architecture with PyTorch.
- Built a structured preprocessing and training pipeline (RGB normalization, 256×256 resizing, Adam optimizer with BCE + Dice loss) and achieved ~0.74 Dice coefficient and ~0.68 IoU on unseen test data.
- Deployed the trained model as an interactive Streamlit web app. This allows for real-time CPU inference, taking less than 300 ms per image. Users can upload images, view masks, and download segmentation outputs.

EDUCATION

D Y Patil School Of Engineering And Technology

Pune,India

Bachelor of Technology - Computer Engineering (SGPA: 9.15)

06/2022 -08/2026

RELEVANT CERTIFICATES

- Google Data Analytics (Google) 
- Data Fundamentals (IBM) 